



VAREN: Vegetation and Asset Recognition for Electrical Network

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The distribution network has a visibility problem

LiDAR provides an accurate representation of the network and VAREN transforms it into operational intelligence.

- **Challenges :**

- Distribution assets are numerous, diverse and spread across large areas
- Vegetation outages account for 45% of the total number of HQ outages
- The traditional GIS system are not designed to support fast and intelligent decision-making

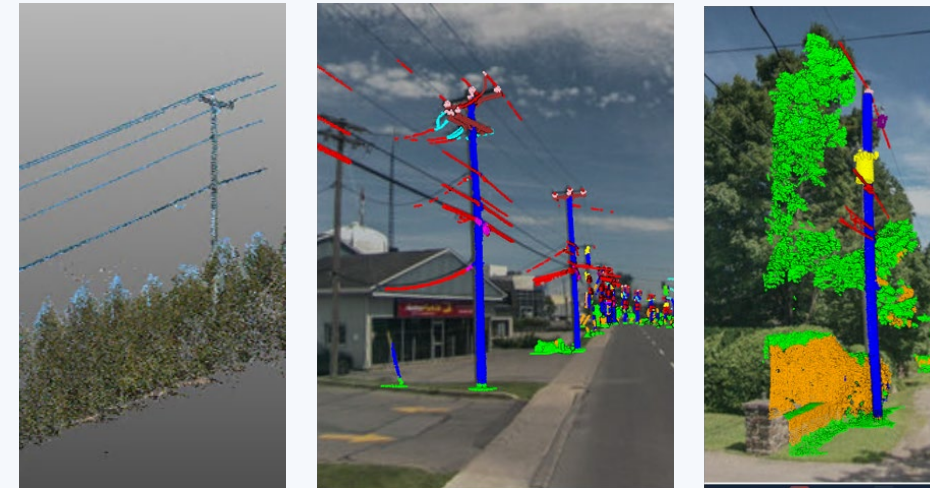
- **Our solution :**

- Mobile LiDAR and imagery are available and powerful enough to have an accurate view of the network

Raw LiDAR
Manual interpretation
Slow corrections

VAREN

Assets
Spans
Clearances
GIS-ready outputs



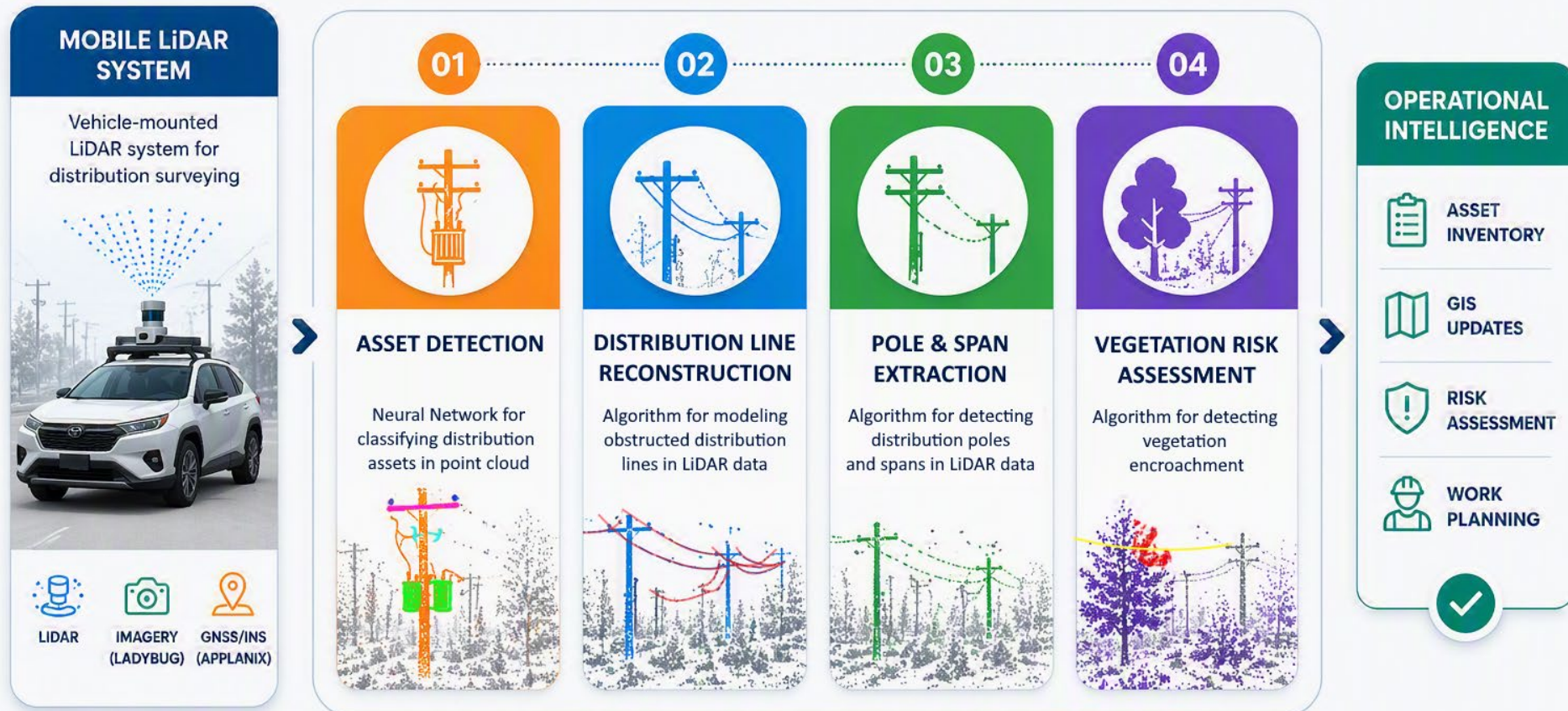
More data ≠ more intelligence

Introducing VAREN

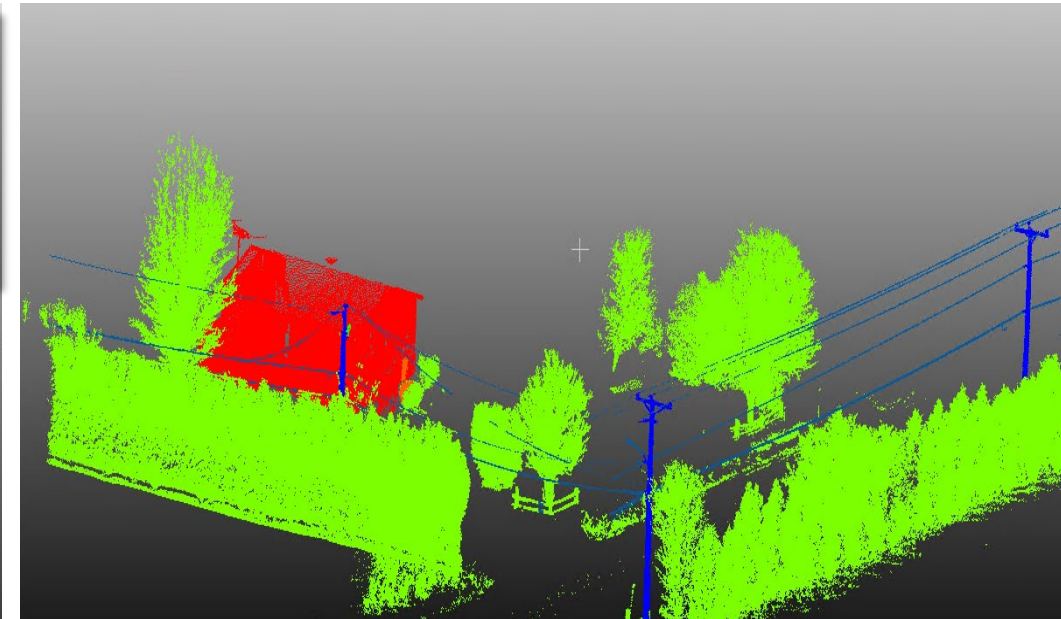
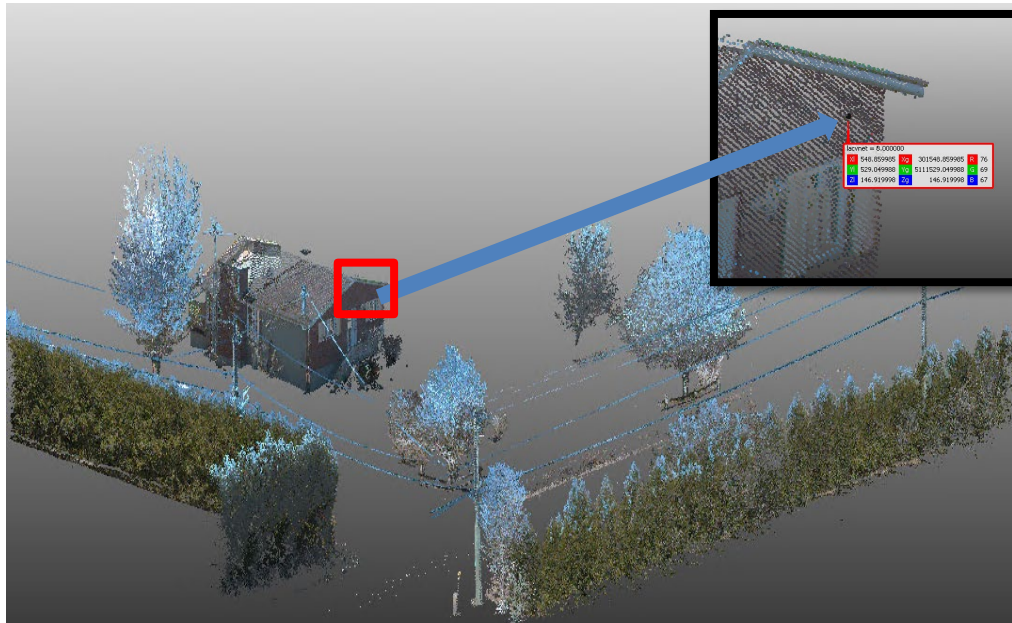
An open-source library to extract actionable intelligence from mobile LiDAR point clouds for Distribution networks.

Vegetation management use case:

— OPEN-SOURCE INTELLIGENCE LAYER —



Neural Network for LiDAR classification



Automatic LiDAR recognition: each points has a label (Houses, Vegetation, Poles, Wires, etc.)

Confusion Matrix of the LiDAR predicted Label

TABLE I
DATASET HQ-2025

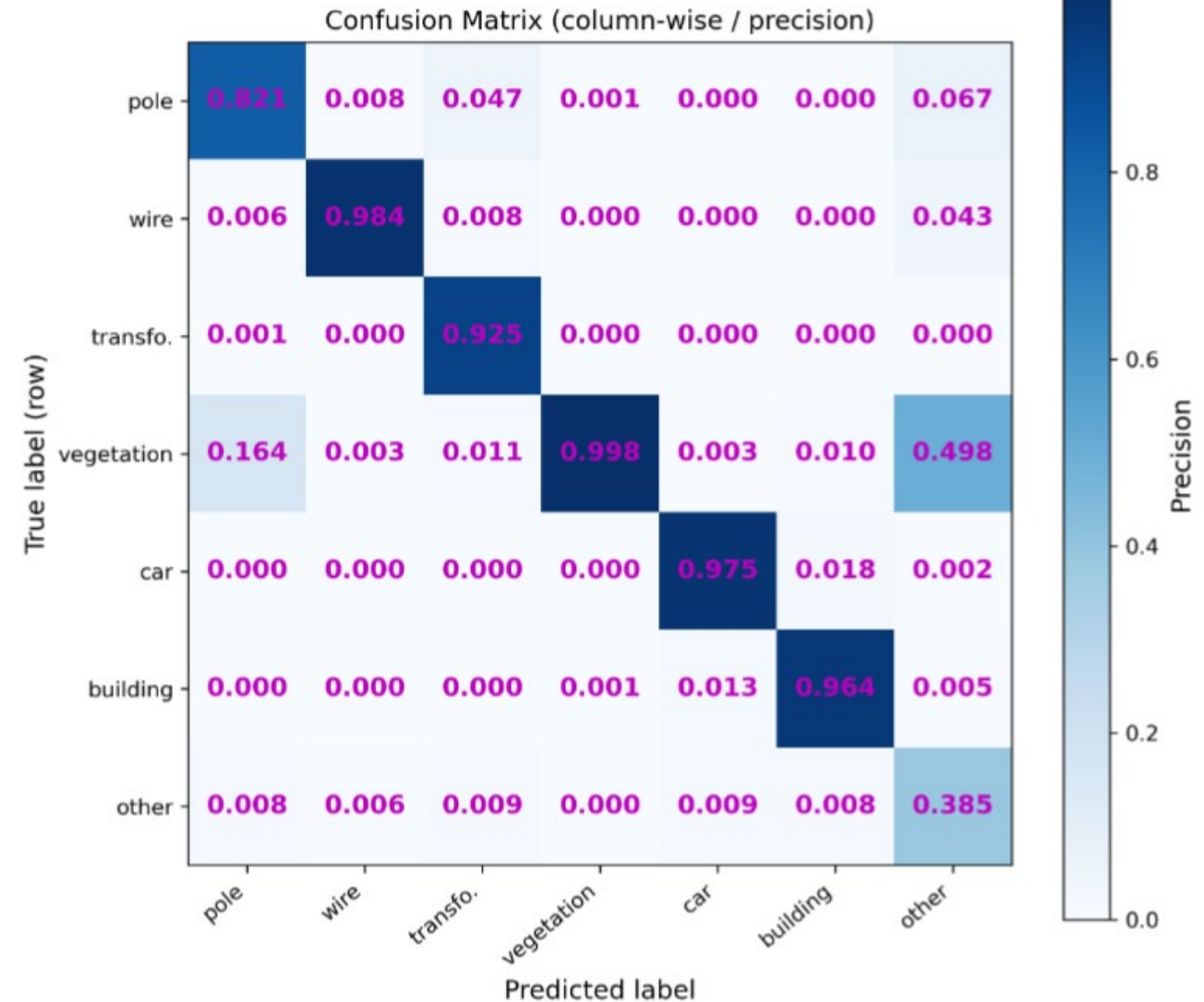
Class	Train (%)	Val (%)	Test (%)	Total Count (M)
Vegetation	89.79	72.54	89.98	134.6
Wire	1.31	3.24	1.54	2.6
Pole	0.98	2.25	1.16	1.9
Transformer	0.04	0.11	0.03	0.07
Building	5.71	17.29	6.26	11.9
Car	1.38	1.69	0.83	2.0
Other	0.79	2.89	0.20	1.5
Total Points	91.0M	24.4M	39.1M	154.5M

VAREN → 154M

VS

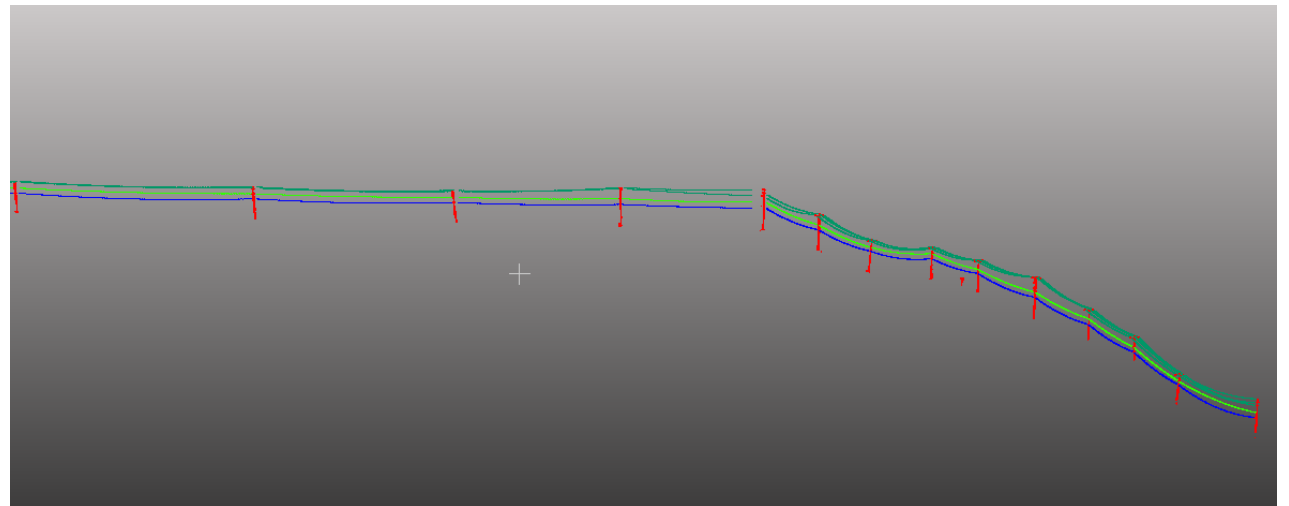
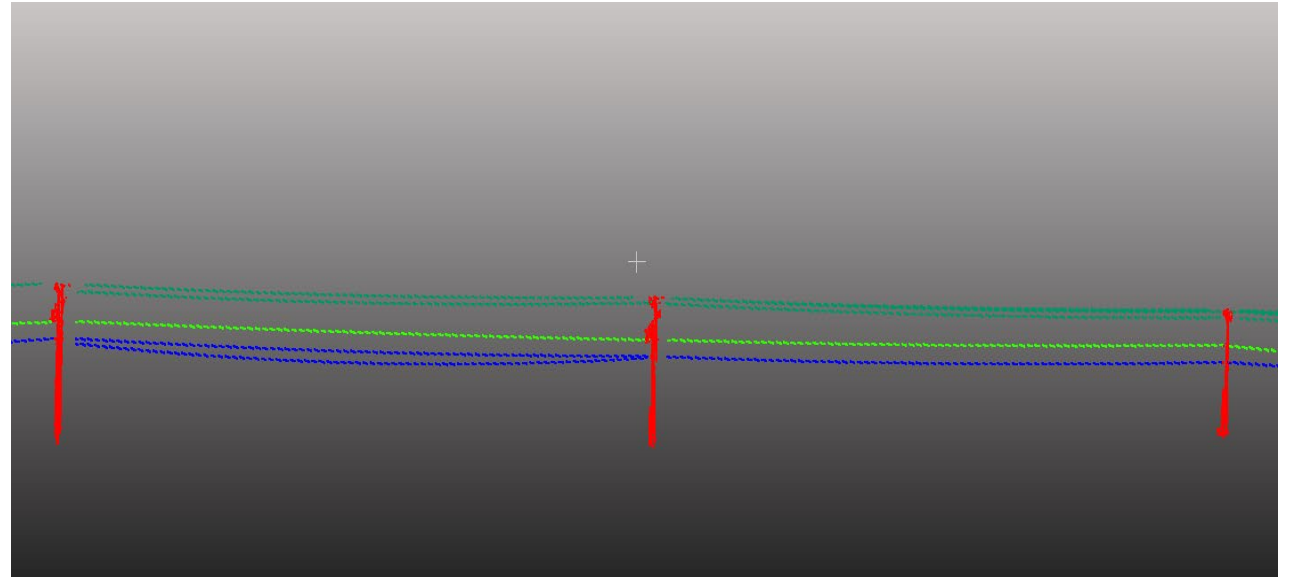
Toronto3D* → 78M points

* (<https://github.com/WeikaiTan/Toronto-3D>)



The MV, LV and Telco power line identification

We extended our previous algorithm by adding a new layer to recognize different line categories: medium voltage, low voltage, and telecommunication.



Power lines detection and reconstruction

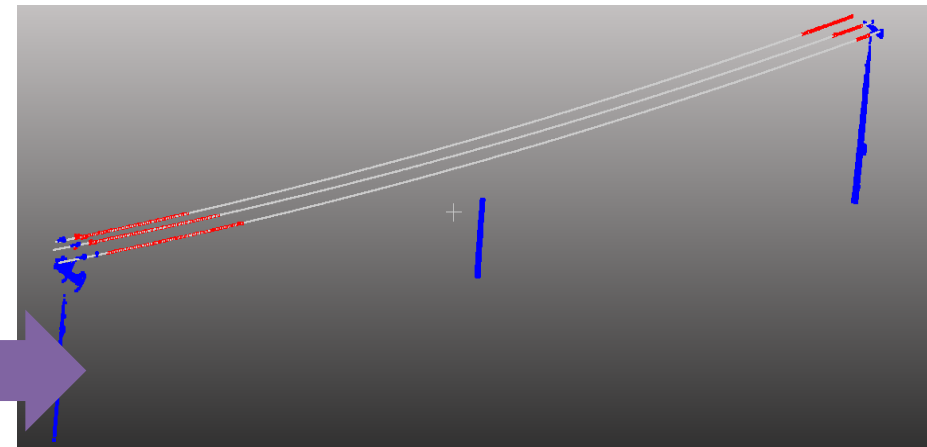
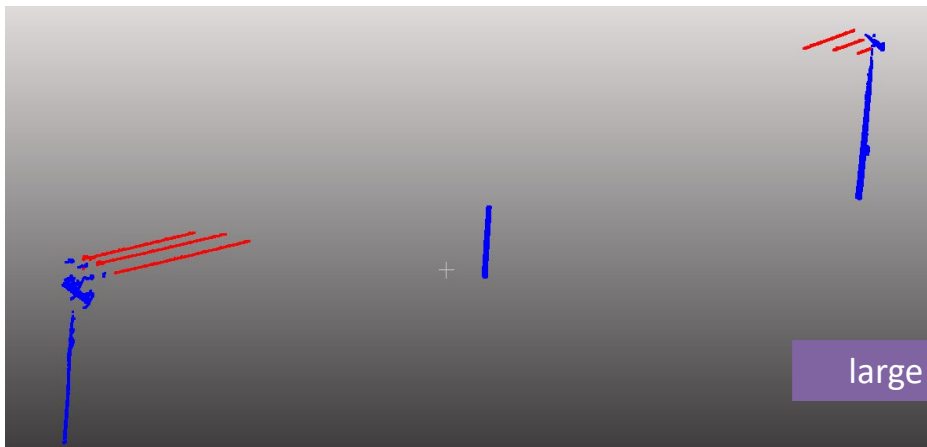
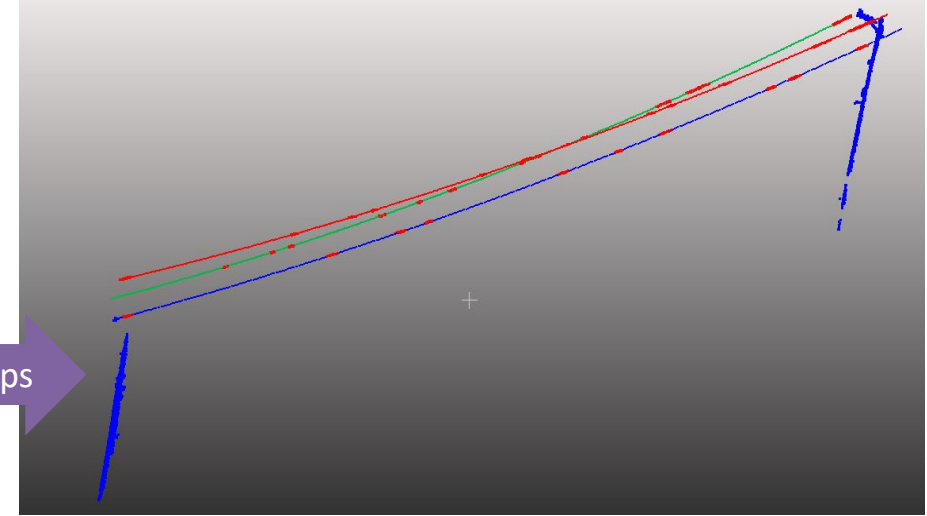
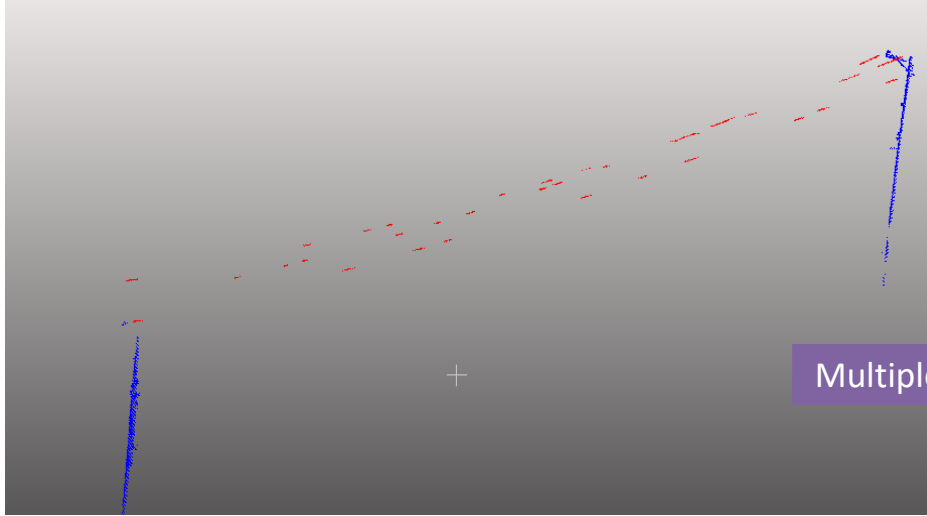
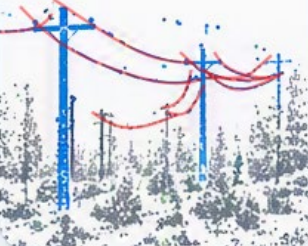
A highly advanced algorithm designed to reconstruct incomplete LiDAR powerline data in the presence of obstructions and low point density.

02



DISTRIBUTION LINE RECONSTRUCTION

Algorithm for modeling obstructed distribution lines in LiDAR data



Performance Indicator & Benchmarking

- Tested on more than 10,870 power spans for a total of **382KM** representing **23,372** powerlines

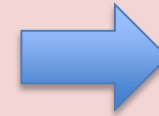


TABLE III: Macro-Level performance table for Aerial LiDAR

Detected	TP	FP	Prec. %	Recall %	F1 %	RMSE
22 554	22 201	550	96.8	94.5	95.6	0.1265

CAO: Di Cao et al. "An Automatic Method for Powerline Extraction From ALS Point Cloud of Powerline Corridors"

EPL: Zhenwei Shi, Yi Lin, and Hui Li. "Extraction of urban power lines and potential hazard analysis from mobile laser scanning point clouds"

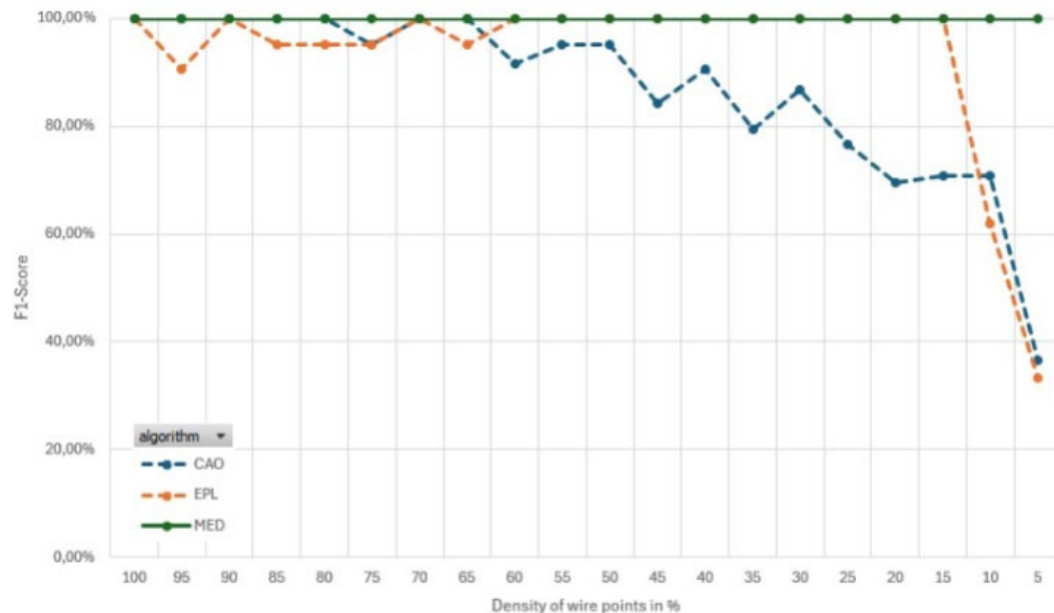


Fig. 9: Comparison of F1-Score between the three powerline reconstruction algorithms for the Small Gaps scenario

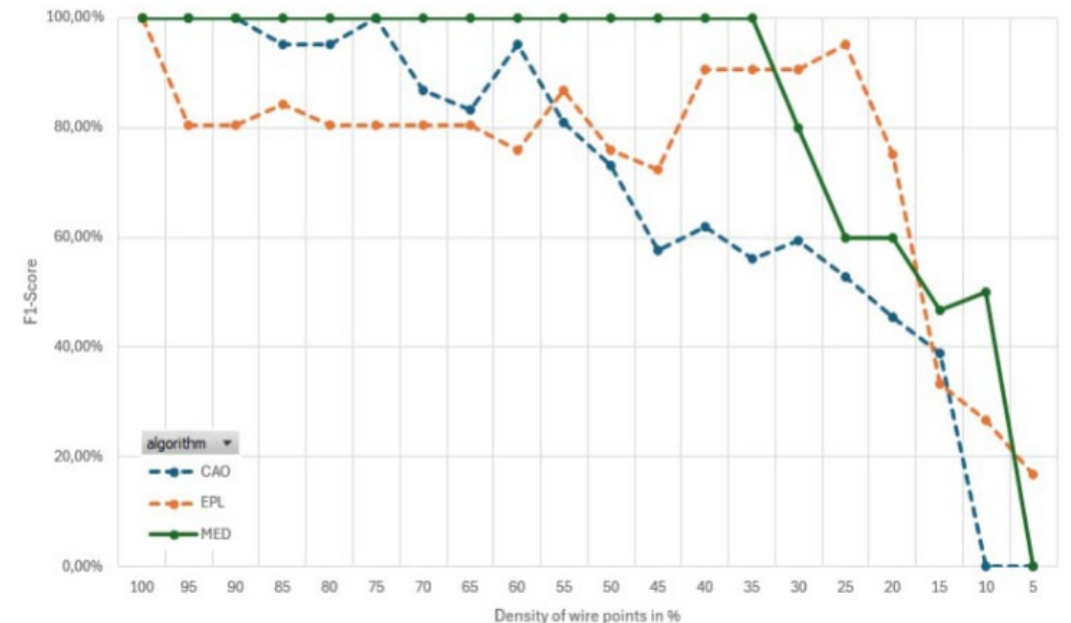
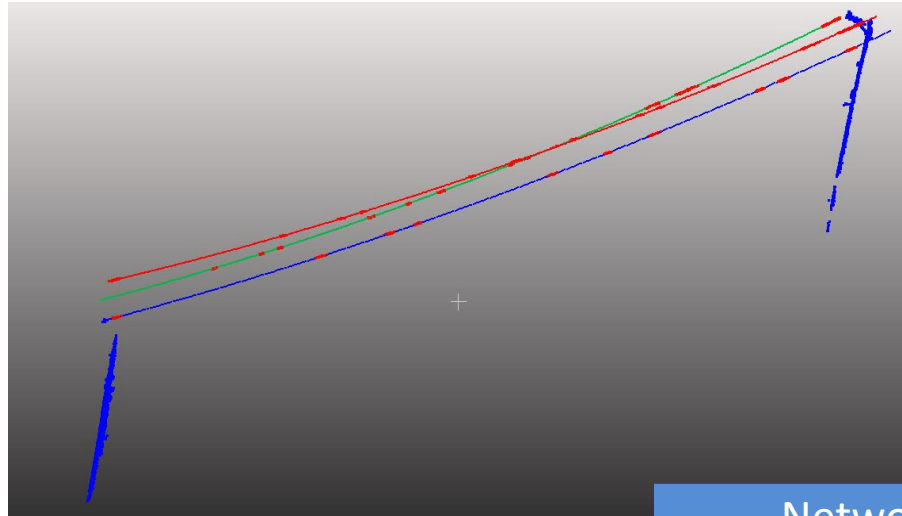
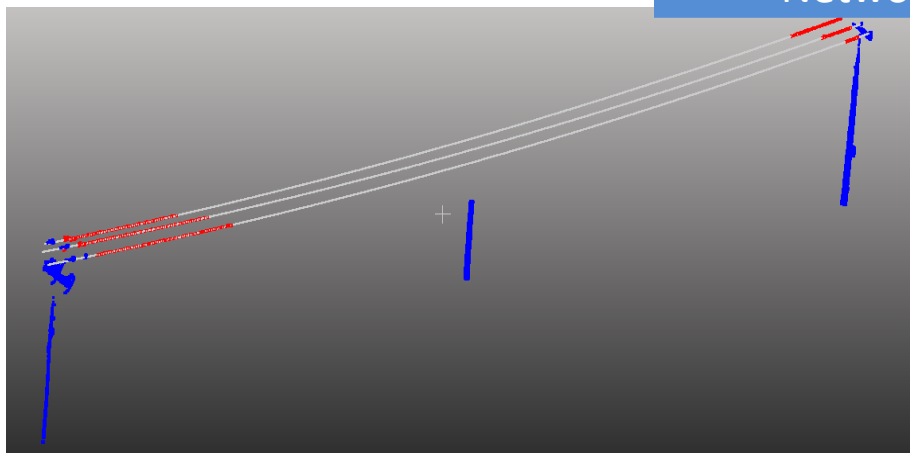


Fig. 7: Comparison of F1-Score between the three powerline reconstruction algorithms for the Main Gap scenario

Combining Poles, Lines and Span to RE-create Distribution Network



Network



- Based on the output of the line reconstruction our algorithm permits to reconstruct the network from labeled LiDAR without any prior information from GIS system

Vegetation encroachment

04

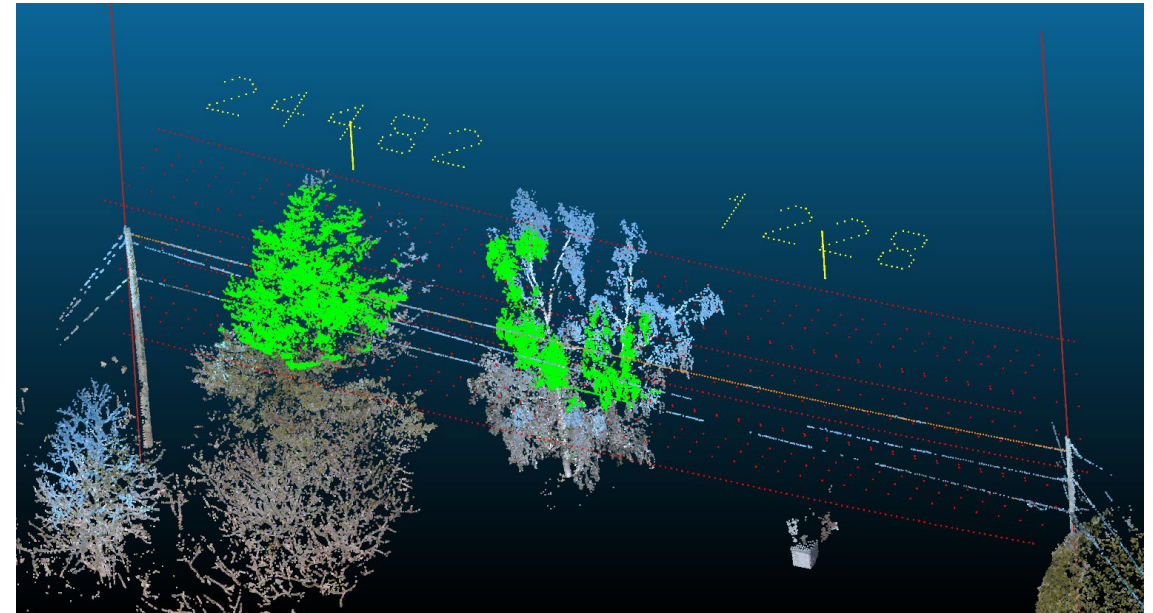
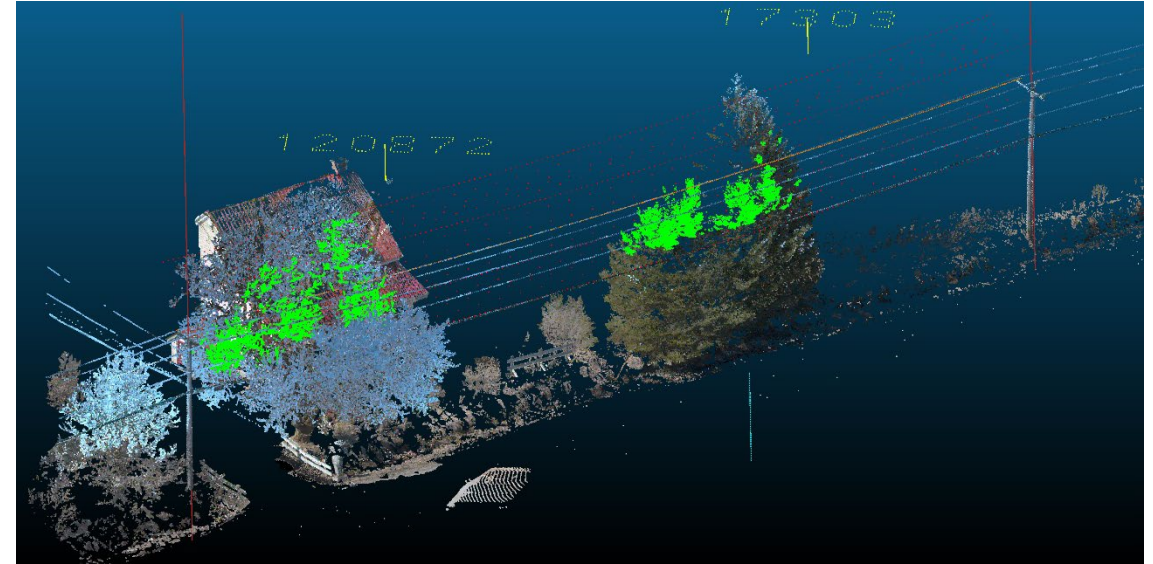
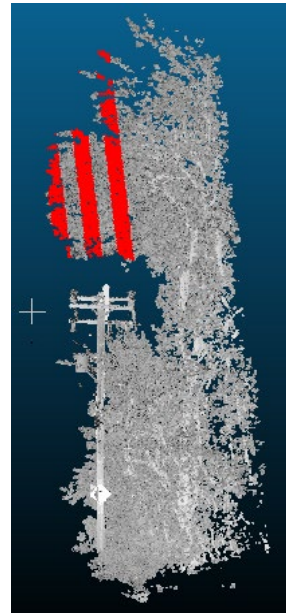
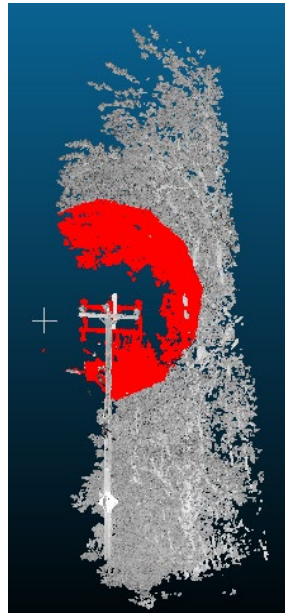


VEGETATION RISK ASSESSMENT

Algorithm for detecting vegetation encroachment



- Generation of automated pruning work order
- Automatic identification of risk zones
- Operational efficiency
- Span accessibility for vegetation crews

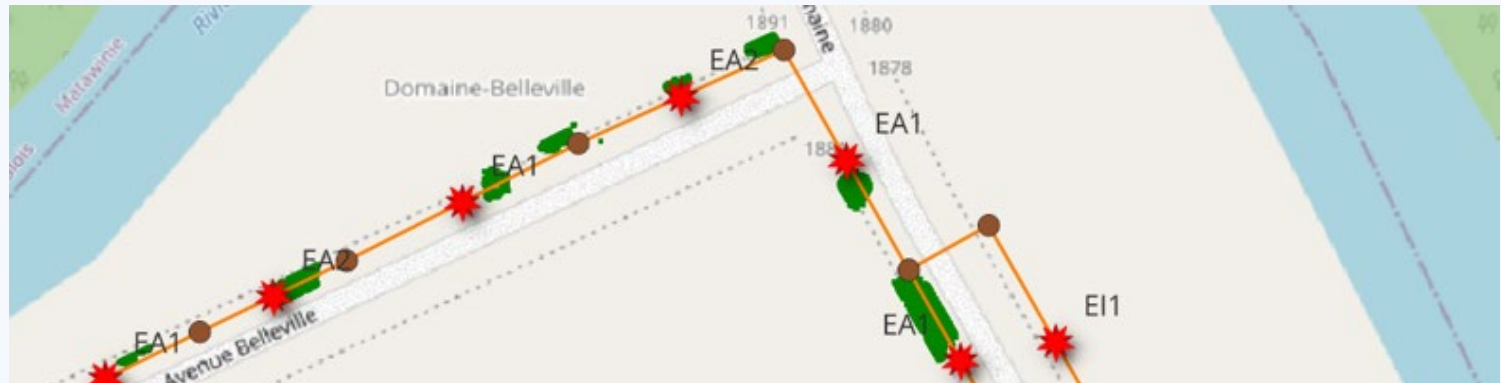
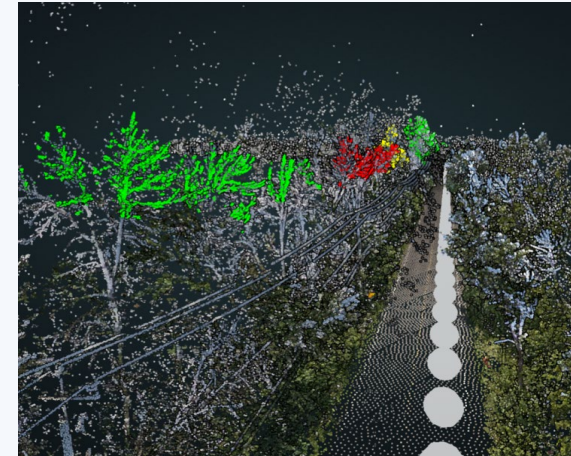
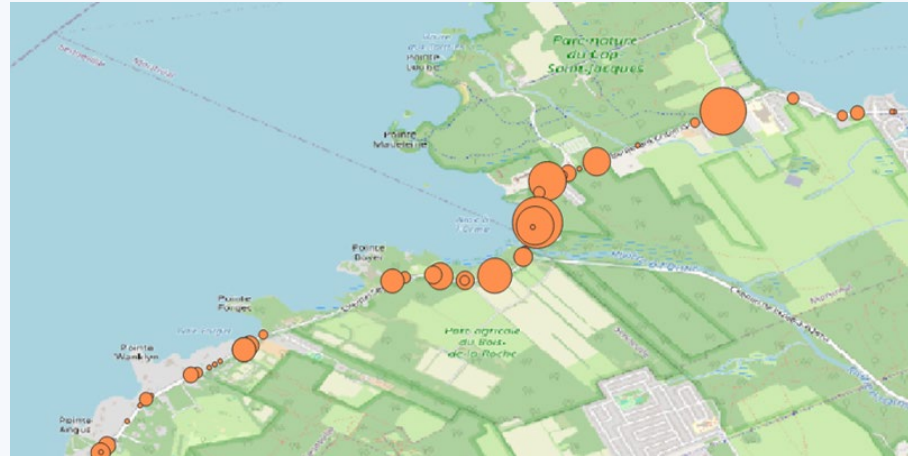
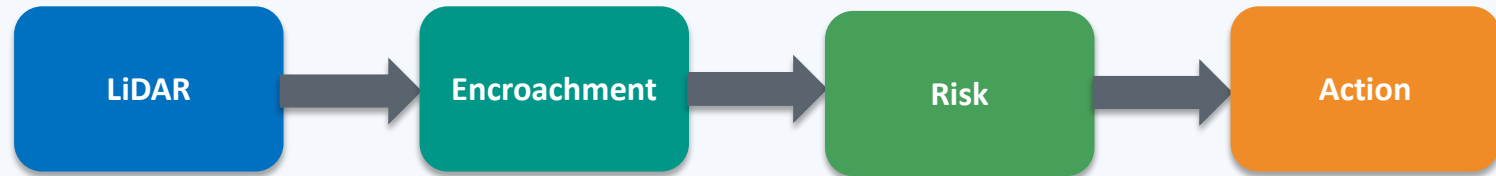


From LiDAR to Vegetation Risk Assessment

VAREN bridges sensing, modelling, and operational action.

Multiple Benefits:

- Asset mapping
- Network reconstruction
- GIS data-quality improvement
- Vegetation risk assessment & prioritization
- Generation of automated pruning WO
- Operational efficiency :
 - Span accessibility
 - Pruning/felling strategy
 - Standard risk assessment
- Data-twin enablement



Why VAREN as a LF Energy project?

Accelerating OSS AI for Utility Infrastructure

Build a shared ecosystem instead of isolated utility-specific solutions to reduce duplication across utilities



MPL-2.0

Enables open collaboration and development while supporting commercial adoption.

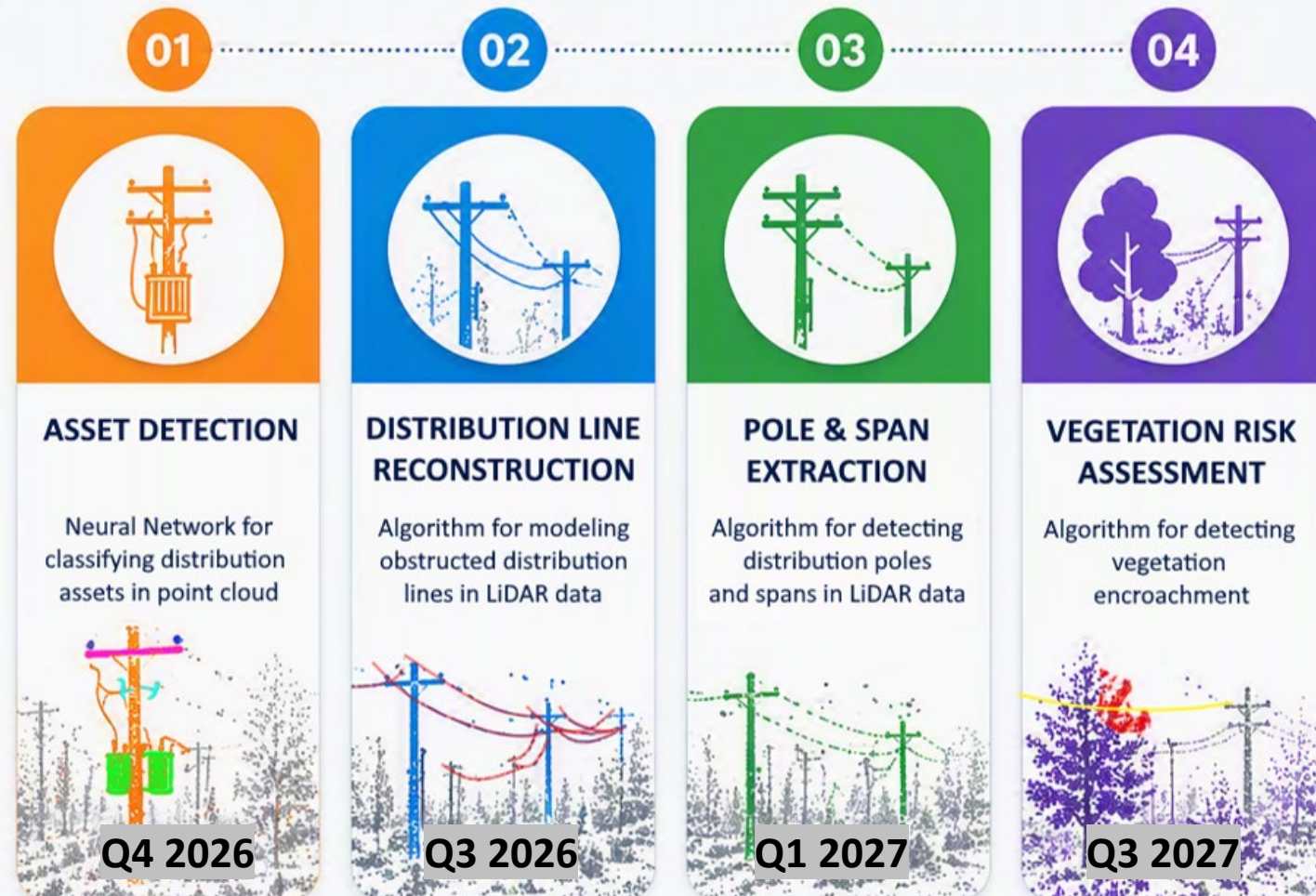
..."creating a technology ecosystem to support rapid decarbonization that benefits the environment, enables economic prosperity, and leads to social well-being for future generations..."

From <https://lfenergy.org/about/why-lf-energy/>

Project Release Roadmap

VAREN is proposed as a new LF Energy Sandbox project.

- VAREN release readiness:
 - Algorithms adapted to LFE Standards
 - Documentation and vetted Data set
 - Initial release upon TAC acceptance
 - Licensing: **MPL-2.0**
 - Language: 
 - Main Libraries: LACV-Net



Thank you very much for your time

GIT repo : [here](#)

